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Question 1: Daily Tasks of a System Administrator

As the system administrator of a small company, my day would begin with monitoring the overall health of all IT systems — checking servers, network infrastructure, and connected devices for any signs of unusual activity or performance issues. Daily tasks include monitoring system health, checking logs for anomalies, responding to user support tickets, and ensuring backups are completed successfully. For example, I would review event logs every morning to catch any unauthorized login attempts or failed services before they escalate into bigger problems. Security best practices such as installing antivirus software, updating virus definitions regularly, configuring firewall rules, and auditing logs are essential to preventing unauthorized access and maintaining system integrity.Additionally, key responsibilities include installing and updating hardware and software, monitoring system performance, troubleshooting issues, implementing security measures, and providing technical support to users all of which contribute to minimizing downtime and keeping the company's operations running smoothly throughout the day.

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Question 2: Troubleshooting Network Permission Issues

When a user reports that they cannot access the network due to permission issues, the first step I would take is to verify their account credentials and confirm that their user account is active and properly configured in the system. System administrators are responsible for managing user access to systems and networks, including handling user permissions, setting up new accounts, and ensuring that all access is properly controlled and secure. Next, I would check the user's group membership and access rights — for instance, in a Windows environment, I would open Active Directory and confirm whether the user belongs to the correct security groups that grant network access. It is also important to double-check the sharing and security permissions on the shared folder or resource, confirm that network discovery is enabled, and verify that all computers are part of the same workgroup or domain. If the permissions appear correct but the issue persists, I would then inspect the Event Viewer logs on both the user's machine and the server to identify specific error codes, and if needed, reset the TCP/IP stack or restart networking services to restore proper connectivity. If the error persists after checking permissions, additional steps such as flushing the DNS cache and checking firewall or antivirus settings may also help resolve the problem.

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Question 3: VirtualBox as a Virtual Laboratory for System Administrators

A Virtual Laboratory using VirtualBox is an invaluable tool for system administrators because it provides a safe and isolated environment to practice real-world tasks without any risk to live production systems. A virtual laboratory is a digital environment that simulates real laboratory tools, systems, or machines using virtualization software, allowing experiments to be done safely without risk and providing hands-on practice without needing expensive hardware. For example, a system administrator in training can set up a virtual domain controller, configure Active Directory, simulate server failures, and practice disaster recovery — all within VirtualBox — without ever touching the actual company infrastructure. VirtualBox can be used to quickly set up a lab environment, and the ability to rapidly provision multiple servers for testing is very useful for those learning system administrator responsibilities, including mimicking production environments that may include application servers, web servers, and databases. Furthermore, virtual machines allow you to run any operating system needed for training, create multiple VMs connected in a private network to practice network security concepts, and use snapshots to go back to a previous state whenever a mistake is made. This snapshot and rollback feature is particularly beneficial because it eliminates the fear of making irreversible errors, making VirtualBox an ideal learning and testing platform for aspiring and current system administrators alike.

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